



2012
HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time - 5 minutes
- Working Time - 2 hours
- Write using a blue or black pen
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11 - 14
- Begin each question on a new page.
- Write your name and your teacher's name on the booklet and your Multiple Choice answer sheet.

Total marks (70)

Section I

10 marks

- Attempt Questions 1-10
- Answer on the Multiple Choice answer sheet provided.
- Allow about 15 minutes for this section

Section II

60 marks

- Attempt questions 11 – 14
- Answer in the booklet provided and show all necessary working.
- Start a new page for each question and clearly label it.
- Allow about 1 hour 45 minutes for this section

Section I**Total marks (10)****Attempt Questions 1-10****Allow about 15 minutes for this section**

Use the multiple choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 = ?$ (A) 2 (B) 6 (C) 8 (D) 9A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:A ☒ B ☒ C ☐ D ☐
correct
↖

1. Find the value of a such that $P(x) = x^3 - 2x^2 - ax + 6$ is divisible by $x + 2$.

(A) -5 (B) -3 (C) 3 (D) 5

2. Find the acute angle (to the nearest degree) between the lines
 $x - y = 2$ and $2x + y = 1$.

(A) 18° (B) 27° (C) 45° (D) 72°

3. Find $\int \frac{dx}{1 + 4x^2}$

(A) $\frac{1}{2} \tan^{-1} 2x + c$

(B) $2 \tan^{-1} 2x + c$

(C) $2 \tan^{-1} \frac{x}{2} + c$

(D) $\frac{1}{2} \tan^{-1} \frac{x}{2} + c$

4. Identify the derivative of $x^2 \cos^{-1} x$.

(A) $\frac{x^2}{\sqrt{1-x^2}} - 2x \cos^{-1} x$

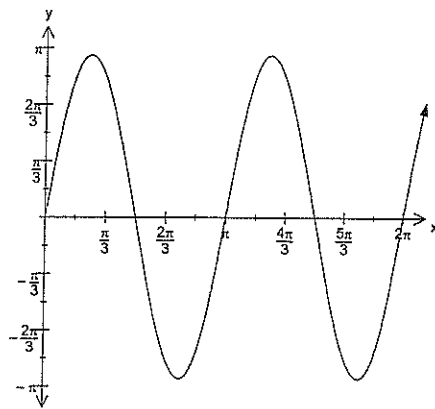
(B) $-\frac{x^2}{\sqrt{1-x^2}} - 2x \cos^{-1} x$

(C) $2x \cos^{-1} x - \frac{x^2}{\sqrt{1-x^2}}$

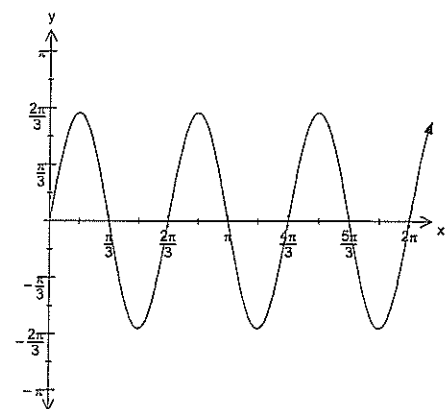
(D) $2x \cos^{-1} x + \frac{x^2}{\sqrt{1-x^2}}$

5. Which graph represents the curve $y = 2\sin 3x$?

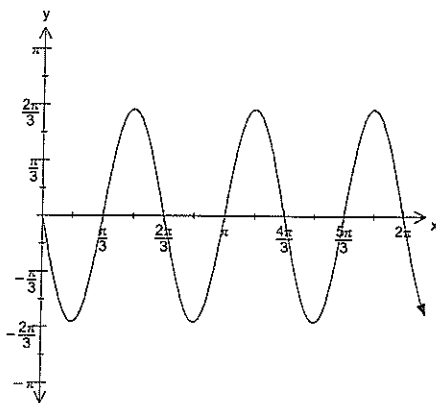
(A)



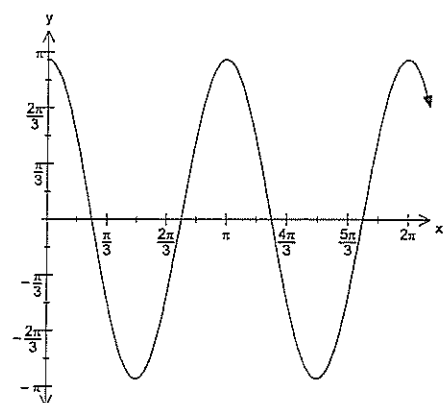
(B)



(C)



(D)



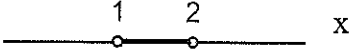
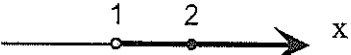
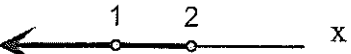
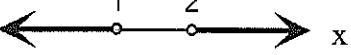
6. If $f(x) = \frac{2}{x+1}$, what is $f^{-1}(x)$?

(A) $y = \frac{x+1}{2}$

(B) $y = \frac{2-x}{x}$

(C) $y = \frac{2-x}{2}$

(D) $y = \frac{2+x}{x}$

7. Given that $\log_a 2 = x$, find an expression for a^{3x} .
- (A) 8
(B) x^{6x}
(C) 2^{3x^2}
(D) a^{3a^2}
8. For what values of x is $\frac{x+4}{x-1} < 6$?
- (A)  x
(B)  x
(C)  x
(D)  x
9. Evaluate $\lim_{x \rightarrow 0} \frac{5x \cos 2x}{\sin x}$
- (A) -10
(B) -5
(C) 5
(D) 10
10. Identify the domain and range of $f(x) = \sin^{-1} 2x$.
- (A) Domain $\rightarrow \{x: -\frac{1}{2} \leq x \leq \frac{1}{2}\}$ and Range $\rightarrow \{y: -\frac{\pi}{4} \leq \sin^{-1} 2x \leq \frac{\pi}{4}\}$
(B) Domain $\rightarrow \{x: -\frac{1}{2} \leq x \leq \frac{1}{2}\}$ and Range $\rightarrow \{y: -\frac{\pi}{2} \leq \sin^{-1} 2x \leq \frac{\pi}{2}\}$
(C) Domain $\rightarrow \{x: -2 \leq x \leq 2\}$ and Range $\rightarrow \{y: -\frac{\pi}{2} \leq \sin^{-1} 2x \leq \frac{\pi}{2}\}$
(D) Domain $\rightarrow \{x: -2 \leq x \leq 2\}$ and Range $\rightarrow \{y: -\pi \leq \sin^{-1} 2x \leq \pi\}$

End of Section 1

Section II**Total marks (60)****Attempt Questions 11 - 14****Allow about 1 hour 45 minutes for this section.**

Answer all questions, starting each question on a new sheet of paper with your name and question number at the top of the page. Do not write on the back of sheets.

Question 11 (15 Marks)	Use a Separate Sheet of paper	Marks
a) Find $\int e^{\frac{x}{4}} dx$		1
b) Find the exact value of $\int_{\frac{3\sqrt{3}}{2}}^3 \frac{2dx}{\sqrt{9-x^2}}$		3
c) Evaluate $\int_0^{\frac{\pi}{4}} \sec^2 x e^{\tan x} dx$ using the substitution $u = \tan x$.		3
d) Find the largest possible domain of $y = \ln(\sin^{-1} x)$.		2
e) Solve for x : $\left(x + \frac{1}{x}\right)^2 - 5\left(x + \frac{1}{x}\right) + 6 = 0$		2
f) AB is the diameter and AC a chord of a circle. The bisector of $\angle BAC$ cuts the circle at D .		
(i) Construct a diagram showing all of this information.		1
(ii) The tangent at D meets AC produced at E . Prove that the tangent is perpendicular to AE .		3

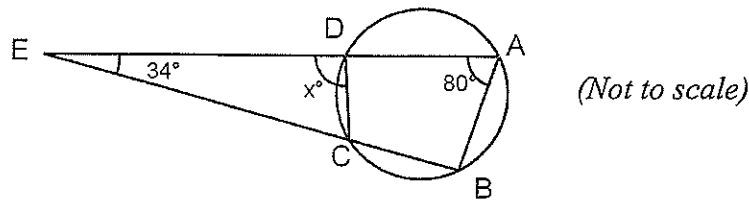
End of Question 11

Question 12 (15 Marks)

Use a separate sheet of paper

Marks

- a) Find the value of x , giving reasons for your answer.

2

- b) i) Without using calculus, sketch $y = (x - 1)(x^2 - 4)$

2

- ii) Hence, solve the inequality $(x - 1)(x^2 - 4) < 0$

1

- c) A spherical balloon leaks air such that the radius decreases at a rate of 0.5 cm s^{-1} . Calculate the rate of change of the volume of the balloon when the radius is 10 cm .

3

- d) Find the exact value of $\int_0^1 \frac{x dx}{1 + x^2}$

2

- e) Evaluate $\lim_{x \rightarrow \infty} \frac{x^2 - 2}{3x^2 - x + 1}$

1

- f) Prove by mathematical induction that (for n a positive integer)

4

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

End of Question 12

Question 13 (15 Marks)

Use a Separate Sheet of paper

Marks

- a) A particle moves such that its displacement x cm from the origin, O after time t seconds is given by:

$$x = \sqrt{3} \cos 3t - \sin 3t$$

- (i) Show that the particle moves in Simple Harmonic Motion. 2
- (ii) Evaluate the period of motion. 1
- (iii) Find the time when the particle first passes through the origin. 2
- (iv) Find the velocity when the particle is 1 cm from the origin on its first oscillation. 2

- b) (i) Prove $\frac{d^2 x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ 1

- (ii) Prove $\frac{d}{dx} (x \ln x) = 1 + \ln x$ 1

- (iii) The acceleration of a particle moving in a straight line and starting from rest at 1 cm on the positive side of the origin is given by:

$$\frac{d^2 x}{dt^2} = 1 + \ln x$$

- (α) Derive the equation relating v and x . 2

- (β) Hence, evaluate v when $x = e^2$. 1

- c) The region bounded by $y = \ln x$, $x = 2$, $x = 5$ and the x -axis is rotated about the x -axis.

- (i) Write an integral expression for the volume formed in terms of y and dx . Do not evaluate this integral. 1

- (ii) Use the trapezoidal rule with four function values (3 strips) to find an approximation to this volume (2 decimal places). 2

End of Question 13

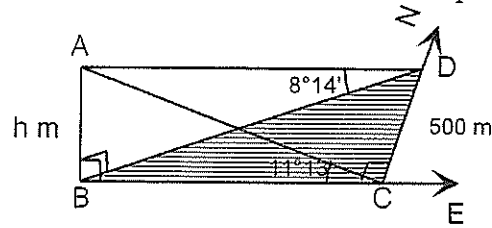
Question 14 (15 Marks)

Use a Separate Sheet of paper

Marks

- a) A is the top of a vertical mast AB standing on level ground. Two points C and D are on horizontal ground such that C is due East of B and D is 500 m due North of C . The angles of elevation of A from C and D respectively are $11^\circ 13'$ and $8^\circ 14'$.

4



Calculate the height, h of the tower to the nearest metre.

- b) The polynomial equation $8x^3 - 36x^2 + 22x + 21 = 0$ has roots which form an arithmetic progression. Find the roots. 3
- c) For the arithmetic sequence $\log_{10}(x-2)$, $\log_{10}(x-2)^2$, $\log_{10}(x-2)^3$, ..., show that the sum of n terms is $\frac{n}{2} \log_{10}(x-2)^{n+1}$. 3
- d) One hundred grams of sugar cane in water are being converted into dextrose at a rate which is proportional to the amount at any time. That is, if M grams are converted in t minutes, then $\frac{dM}{dt} = k(100 - M)$ where k is a constant.
- (i) Show that $M = 100 + Ae^{-kt}$, where A is a constant, satisfies the differential equation. 2
- (ii) Find A , given that when $t = 0$, $M = 0$. 1
- (iii) If 40 grams are converted in the first 10 minutes, find how many grams are converted in the first 30 minutes. 2

End of Examination

STHS Trial HSC Examination – Mathematics Extension 1 2012

Multiple Choice Answer Sheet

Name _____

Completely fill the response oval representing the most correct answer.

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

Multiple Choice

- ① D ③ A ⑤ B ⑦ A ⑨ C
 ② D ④ C ⑥ B ⑧ D ⑩ B

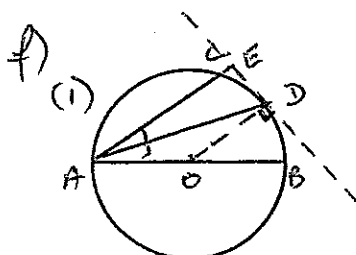
Q11 a) $\int e^{\frac{x}{4}} dx = 4 e^{\frac{x}{4}} + C$

b) $\int_{\frac{\sqrt{3}}{2}}^3 \frac{2 dx}{\sqrt{9-x^2}} = \left[2 \sin^{-1} \frac{x}{3} \right]_{\frac{\sqrt{3}}{2}}^3$
 $= \frac{\pi}{3}$

c) $\int_0^{\frac{\pi}{4}} \sec^2 x e^{\tan x} dx = u = \tan x$
 $\therefore \frac{du}{dx} = \sec^2 x$
 $\therefore du = \sec^2 x dx$
 $= \int_0^1 e^u du$
 $= e - 1$

d) Domain of $\sin^{-1} x \Rightarrow -1 \leq x \leq 1$
 Domain of $\ln x \Rightarrow x > 0$
 $\therefore$ Domain of $\ln(\sin^{-1} x) \Rightarrow 0 < x \leq 1$

e) $(x + \frac{1}{x})^2 - 5(x + \frac{1}{x}) + 6 = 0$
 let $u = x + \frac{1}{x}$
 $\therefore u^2 - 5u + 6 = 0$
 $\therefore (u-2)(u-3) = 0$
 $\therefore x + \frac{1}{x} = 2 \quad \text{or} \quad x + \frac{1}{x} = 3$
 $\therefore x^2 - 2x + 1 = 0 \quad x^2 - 3x + 1 = 0$
 $\therefore (x-1)^2 = 0 \quad \therefore x = 3 \pm \sqrt{9-4}$
 $\therefore x \geq 1$
 $= \frac{3 \pm \sqrt{5}}{2}$



Dotted lines are constructions for part (ii)

- (ii) Construct: OD (radius)
 Extend AC to E
 on tangent.

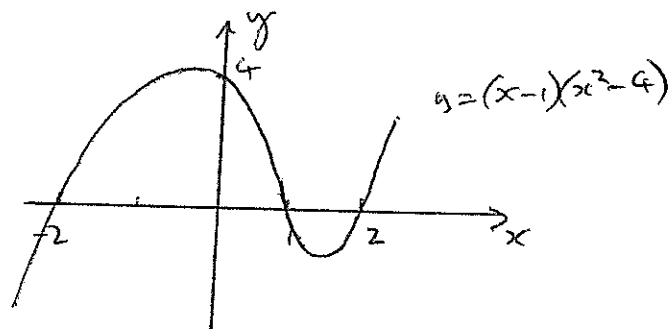
Proof: $\triangle OAD$ is isosceles (OA and OD radii)
 $\angle OAD = \angle ODA$ (data)
 $\angle ODA = \angle DAE$ (base $\angle$ s of isosceles $\triangle$)
 $\therefore \angle OAD = \angle DAE$
 $\therefore AE \parallel OD$ (alternate $\angle$ s)
 $OD \perp DE$ (radius at point of contact $\perp$ tangent)
 $\therefore AE \perp DE$ QED

(Not the only approach)
 e.g. using alternate angle

Q 12

- a) $\angle C + \angle A = 100^\circ$ (opposite $\angle$ s in cyclic quadrilateral are supplementary)
 $\therefore x + 74 = 100$ (external $\angle$ equal to sum of opposite internal $\angle$ s)
 $\therefore x = 66^\circ$

b) (i) $y = (x-1)(x^2-4)$
 $= (x-1)(x-2)(x+2)$
 leading coefficient $= 1 > 0$



(ii) $(x-1)(x^2-4) < 0$
 $\therefore x < -2, 1 < x < 2$

c) $\frac{dr}{dt} = -0.5 \text{ cm s}^{-1}$

Now $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt}$

$= 4\pi r^2 \times -0.5$

$= -2\pi r^2$

$= -200\pi \text{ cm}^3 \text{ s}^{-1}$

$V = \frac{4}{3}\pi r^3$
 $\therefore \frac{dV}{dt} = 4\pi r^2$

d) $\int_0^1 \frac{x dx}{1+x^2}$

$= \frac{1}{2} \int_0^1 \frac{2x dx}{1+x^2}$

$= \frac{1}{2} [\ln(1+x^2)]_0^1$

$= \frac{1}{2} (\ln 2 + \ln 1)$

$= \frac{\ln 2}{2}$

e) $\lim_{x \rightarrow \infty} \frac{x^2-2}{3x^2-x+1}$

$= \lim_{x \rightarrow \infty} \frac{1 - \frac{2}{x^2}}{3 - \frac{1}{x} + \frac{1}{x^2}}$

$= \frac{1}{3}$

f) RTS $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$

For $n=1$,

LHS $= \frac{1}{1 \cdot 2}$
 $= \frac{1}{2}$

RHS $= \frac{1}{1+1}$
 $= \frac{1}{2}$
 $= \text{LHS}$

$\therefore$ true for $n=1$.

Assume true for $n=k$.

$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$

For $n=k+1$,

We'd expect $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} = \frac{k+1}{k+2}$

LHS $= \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)}$
 $= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)}$
 $= \frac{k(k+2) + 1}{(k+1)(k+2)} = \frac{k^2 + 2k + 1}{(k+1)(k+2)}$
 $= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2} = \text{RHS}$

$\therefore$ If true for $n=k$, then true for $n=k+1$.

Now,

True for $n=k=1$

$\therefore$ true for $n=k+1=2$

True for $n=k=2$

then true for $n=k+1=3, \dots$

$\therefore$ true for all

integral values of $n, n \geq 1$

Q13

a) (i) $x = \sqrt{3} \cos 3t - \sin 3t$

$$\frac{dx}{dt} = -3\sqrt{3} \sin 3t - 3 \cos 3t$$

$$\frac{d^2x}{dt^2} = -9\sqrt{3} \cos 3t + 9 \sin 3t$$

$$= -9(\sqrt{3} \cos 3t - \sin 3t)$$

$$= -9x$$

$\therefore$ SHM with $n^2 = 9$

(ii) Period of motion $T = \frac{2\pi}{n}$

$$= \frac{2\pi}{3}$$

b) (i) RTs $\frac{d^2x}{dt^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dv} \left(\frac{1}{2} v^2 \right) \cdot \frac{dv}{dx}$$

$$= v \frac{dv}{dx}$$

$$= \frac{dx}{dt} \cdot \frac{dv}{dx}$$

$$= \frac{dv}{dt}$$

$$= \frac{d^2x}{dt^2} \text{ QED}$$

(ii) $\frac{d}{dx} (x \ln x)$

let $u = x$

$$\frac{du}{dx} = 1$$

$$v = \ln x$$

$$\frac{dv}{dx} = \frac{1}{x}$$

$$= v u' + u v'$$

$$= \ln x \cdot 1 + x \cdot \frac{1}{x}$$

$$= \ln x + 1$$

(iii) a) $\frac{d^2x}{dt^2} = 1 + \ln x$

$$\therefore \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 1 + \ln x$$

$$\therefore \int \frac{d}{dx} \left(\frac{1}{2} v^2 \right) dx = \int (1 + \ln x) dx$$

$$\therefore \frac{1}{2} v^2 = \int \frac{d}{dx} (x \ln x) dx$$

$$= x \ln x + C$$

$$\therefore v^2 = 2x \ln x + C$$

When $x=1, v=0 \therefore C=0$

$$\therefore v^2 = 2x \ln x$$

⑥ When $x = e^2, v^2 = 2e^2 \ln e^2 = 4e^2 \therefore v = 2e \text{ cm s}^{-1}$

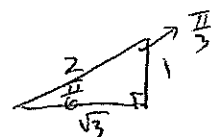
(iii) $\sqrt{3} \cos 3t - \sin 3t = 0$

$$\therefore \sqrt{3} - \tan 3t = 0$$

$$\therefore \tan 3t = \sqrt{3}$$

$$\therefore 3t = \frac{\pi}{3}$$

$$\therefore t = \frac{\pi}{9} \text{ s}$$



(iv) Let $\sqrt{3} \cos 3t - \sin 3t = R \cos(3t + \alpha)$

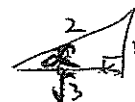
$$= R(\cos 3t \cos \alpha - \sin 3t \sin \alpha)$$

$$\therefore R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = 1$$

$$\therefore \tan \alpha = \frac{1}{\sqrt{3}}$$

$$\therefore \alpha = \frac{\pi}{6}$$



Now, let $2 \cos(3t + \frac{\pi}{6}) = 1$

$$\therefore \cos(3t + \frac{\pi}{6}) = \frac{1}{2}$$

$$\therefore 3t + \frac{\pi}{6} = \frac{\pi}{3}$$

$$\therefore 3t = \frac{\pi}{6}$$

$$\therefore t = \frac{\pi}{18} \text{ s}$$

When $t = \frac{\pi}{18} \text{ s}$,

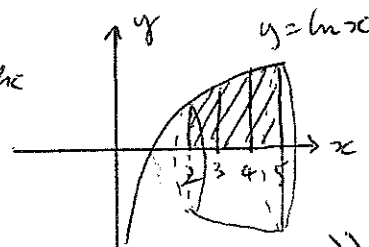
$$v = -3\sqrt{3} \sin \frac{\pi}{6} - 3 \cos \frac{\pi}{6}$$

$$= -3\sqrt{3} \cdot \frac{1}{2} - 3 \cdot \frac{\sqrt{3}}{2}$$

$$= -3\sqrt{3} \text{ cm s}^{-1}$$

c) (i) $V = \pi \int_2^5 (\ln x)^2 dx$

$$= \pi \int_2^5 y^2 dx$$



(ii) $V \approx \pi \frac{1}{2} ((\ln 2)^2 + (\ln 5)^2) + 2((\ln 3)^2 + (\ln 4)^2)$

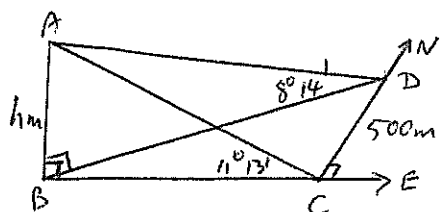
$$= \frac{\pi}{2} [\ln^2 2 + \ln^2 5 + 2\ln^2 3 + 2\ln^2 4]$$

$$\approx 14.6528 \dots$$

$$\approx 14.65 \text{ u}^3 \text{ to 2 dec. pl.}$$

Q14

a)



$$\text{Now } BD^2 - BC^2 = 500^2$$

$$\therefore \frac{h^2}{\tan^2 8^\circ 14'} - \frac{h^2}{\tan^2 11^\circ 13'} = 500^2$$

$$\therefore h^2 \left(\frac{1}{\tan^2 8^\circ 14'} - \frac{1}{\tan^2 11^\circ 13'} \right) = 500^2$$

$$\therefore h^2 = \frac{500^2}{\left(\frac{1}{\tan^2 8^\circ 14'} - \frac{1}{\tan^2 11^\circ 13'} \right)}$$

$$\approx 11193.7458$$

$$\therefore h = 105.8$$

$$\approx 106 \text{ m}$$

$$\begin{aligned} \text{c) } a &= \log_{10}(x-2) \\ d &= \log_{10}(x-2)^2 - \log_{10}(x-2) \\ &= \log_{10}(x-2) \end{aligned}$$

$$\begin{aligned} \therefore S_n &= \frac{n}{2} [2 \log_{10}(x-2) + (n-1) \log_{10}(x-2)] \\ &= \frac{n}{2} [(n+1) \log_{10}(x-2)] \end{aligned}$$

$$= \frac{n}{2} [\log_{10}(x-2)^{n+1}] \quad \text{Q.E.D.}$$

$$\begin{aligned} \text{d) (i) } M &= 100 + A e^{-kt} \\ \frac{dM}{dt} &= -Ak e^{-kt} \\ &= k(100 - (100 + A e^{-kt})) \end{aligned}$$

$$\therefore \frac{dM}{dt} = k(100 - M).$$

When $t=30$,

$$\begin{aligned} M &= 100 - 100 e^{-30 \times 0.05108} \\ &= 78.4 \text{ gm} \end{aligned}$$

$$\frac{h}{BD} = \tan 8^\circ 14'$$

$$\therefore BD = \frac{h}{\tan 8^\circ 14'}$$

$$\text{Also } BC = \frac{h}{\tan 11^\circ 13'}$$

$$\text{b) } 8x^3 - 36x^2 + 22x + 21 = 0$$

Let roots be $x-d, x, x+d$

$$\text{Now } 2x = 3d = \frac{36}{8}$$

$$\therefore d = \frac{3}{2}$$

$$\pi d = d(d^2 - d^2)$$

$$= \frac{3}{2} \left(\frac{9}{4} - d^2 \right)$$

$$= -\frac{21}{8}$$

$$\therefore \frac{9}{4} - d^2 = -\frac{7}{4}$$

$$\therefore d^2 = 4$$

$$\therefore d = \pm 2$$

 $\therefore$ roots are $-\frac{1}{2}, \frac{1}{2}, 3\frac{1}{2}$

$$\begin{aligned} \text{(ii) When } t=0, M=0 \\ \therefore 0 &= 100 + A e^0 \\ \therefore A &= -100 \end{aligned}$$

$$\text{(ii) } M = 100 - 100 e^{-kt}$$

When $t=10$, $M=40$

$$\therefore 40 = 100 - 100 e^{-10k}$$

$$\therefore 100 e^{-10k} = 60$$

$$\therefore e^{-10k} = 0.6$$

$$\therefore \ln e^{-10k} = \ln 0.6$$

$$\therefore -10k = \ln 0.6$$

$$\therefore k = \frac{\ln 0.6}{-10}$$

$$\therefore k \approx 0.05108$$